Powering the UTWind VAWT

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Project Background

Commercial generators are optimized for high speeds, which limits **Vertical Axis Wind Turbine (VAWT)** performance through **high starting torque** and **reduced efficiency at 100-200 RPM.** This work develops a **custom generator** for the **UTWind 2026 turbine**, optimized for **low starting torque** and **improved efficiency at low RPM**. It establishes a **proof-of-concept** for future in-house generator development.

Key Term:
Starting Torque: Minimum torque to overcome startup resistance

Design Methodology

1 ***Brainstorming***
- Four architectures were evaluated.

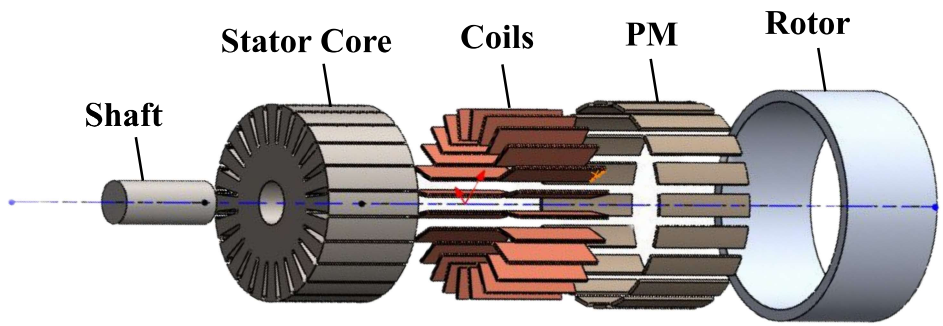
2 ***Decision Matrix***
- Induction & SRG eliminated - poor low-RPM efficiency and high torque ripple.
- AFPMG eliminated - complex manufacturing and poor cogging mitigation.

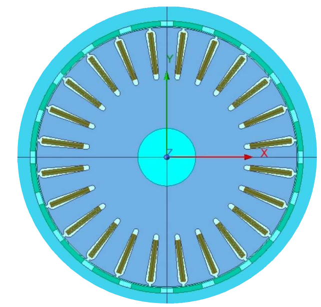
3 ***Optimization Tables - Morph Chart***
- Swept slot/pole, arc ratio, air gap, and pole segmentation

4 ***FEA Validation***
- Developed an initial analytical model using ANSYS RMxprt
- Validated electromagnetic performance in Maxwell 2D

Design Overview

Direct-drive Outer-Rotor Radial-Flux Permanent Magnet Synchronous Generator (RF-PMSG)

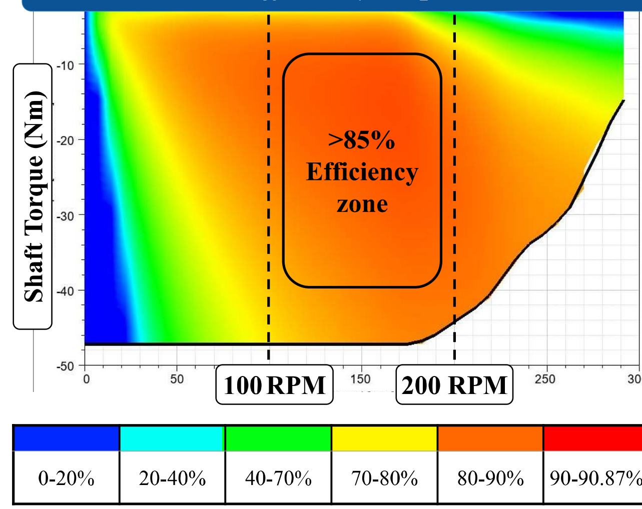




Key Geometry

- ***Slot / Pole***: 24/20
- ***Axial Length:*** 100mm
- ***Air Gap:*** ~1.0mm baseline
- ***Rotor OD:*** ~210 mm
- ***Winding:*** 3-Phase fractional slot concentrated winding (FSCW)
- ***Magnets***: Segmented surface mounted NdFe35

Efficiency Map



Design Goals

0.1 Nm Starting Torque	**600-1200W** Power Output
100-200 RPM Operating Speed	**>90%** Efficiency

Performance

0.09 Nm Peak Starting Torque	**842.8 W** Net Output Power
100-200 RPM Operating Speed	**67.06%** Efficiency

Next Steps

1 ***Phase 1 - Procurement***
Source NdFe35 magnets & ASTM A677 (M19) lamination steel for rotor/stator fabrication

2 ***Phase 2 - Validation***
Guided rotor-stator mating to maintain 1.0mm air gap; verify 600–1200W/0.1 N·m